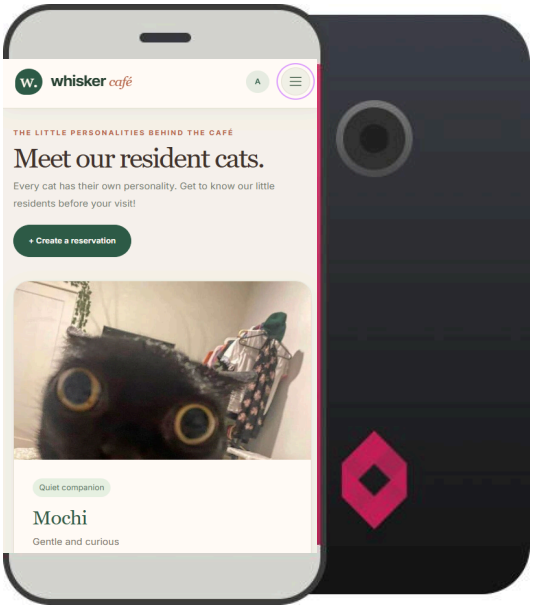
WHISKER CAFE

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Queue and linked list

Explanation:

The **FIFO Queue** follows the First In, First Out principle. This means the first visitor added to the queue is also the first one to be served. In our Whisker Café system, the queue helps organize visitors and manage their turn fairly. We use Python's `deque` to store the queue. The `enqueue` operation adds a visitor to the back, while `dequeue` removes the visitor at the front. Both operations take **O(1) time** because they do not require shifting the other elements.



The **Singly Linked List** stores data using individual nodes. Each node contains a value and a reference to the next node. It uses a `head` to identify the first node and a `tail` to identify the last node. The `append` operation adds a new node at the end. It takes **O(1) time** because the tail directly points to the last node, allowing the new node to be attached without searching through the entire list. On the other hand, the `get` operation takes **O(n) time** because it may need to start at the head and visit each node one by one until it reaches the requested position. Therefore, appending is faster because the last node is directly accessible, while getting an item by position requires traversal.

Group Members and Contributions

Member	Name	Contribution
Member 1	Jenica Vallente	Developed the FIFO Queue implementation and helped integrate the visitor queue into the system.
Member 2	Kent Sungahid	Developed the Singly Linked List implementation and tested its operations.
Member 3	Psalm Walso	Worked on the Django backend, database models, and system functionality.
Member 4	Anilynn Pangca	Designed and developed the website interface, styling, and responsive layout.
Member 5	Daisy Mantalaba	Worked on testing, documentation, screenshots, and project presentation.